

Chinmay Joshi

DATA SCIENCE & DEEP LEARNING RESEARCH
Saarbrücken, Germany · Open to relocating to Amsterdam

chinmay.joshi23@outlook.com | [+49 1788266100](tel:+491788266100) | chinmayj23.github.io | [linkedin/chinmay-joshi23](https://linkedin.com/in/chinmay-joshi23) |

Education

Saarland University

Saarbrücken, Germany

M.Sc. Computer Science, Grade 1.7 (German scale, 1.0 highest)

- **Master's Thesis**, carried out at **CISPA Helmholtz Center for Information Security** (advisor: Dr. Rebekka Burkholz): **On the Effect of Data Pruning on Memorization**. Studied how dynamic data pruning (a training-efficiency method) changes memorization dynamics in deep networks and affects model privacy; proposed **WLIB**, a data-pruning variant that **reduced privacy risk** by 26% on ImageNet with **negligible loss in performance** and **faster training**. [[thesis](#)]
- Recipient of the **SaarlandStipendium** and **Santander Scholarship**.

Narsee Monjee Institute of Management Studies

Mumbai, India

B.Tech. Data Science, GPA 3.8/4.0

Publications

- **Chinmay Joshi**, Advait Gadhikar, Celia Rubio-Madrigal, Aneet Kumar Dutta, Mridula Singh, and Rebekka Burkholz. “Prune to Protect: Faster Training and Enhanced Privacy by Dynamic Data Pruning.” Accepted at the **ICML 2026 Workshop** on Memorization in Foundation Models (MemFM); under review at **NeurIPS 2026**. [[poster](#)]
- **Chinmay Joshi**, David A. Selby, Ayush Patnaik, Sebastian Vollmer, and Gerrit Großmann. “DOLPHIN: Interpretable Discovery of Exceptional Subgroups in Longitudinal Data.” Under review at the NFMCP 2026 Workshop held at ECML-PKDD. [[code](#)]
- **Chinmay Joshi** and Siba Panda. “PCA-LSTM: Deep Learning Approach for the Indian Large-Caps.” 2022 IEEE 7th International Conference for Convergence in Technology (I2CT). [[paper](#)]

Experience

Research Assistant

Dec 2025 – Present

German Research Center for Artificial Intelligence (DFKI)

Kaiserslautern, Germany

- Developing DOLPHIN, an **interpretable subgroup-discovery method** for longitudinal economic data, evaluated on large-scale datasets spanning up to 9,300 entities and 281,000 records (World Bank GDP-per-capita and CMIE household-income surveys).
- Built the pipeline that applies temporal feature engineering, scores each trajectory by its deviation from the population baseline, and selects a compact random forest whose short decision paths translate into **human-readable rules** for the exceptional subgroups.

Research Assistant

Mar 2025 – Sep 2025

Fraunhofer IZFP, Project [Urbanist](#)

Saarbrücken, Germany

- Collected and **processed data streams from mobile and stationary environmental sensor stations**, computing exposure metrics such as UV index and noise levels from aggregated sensor readings.
- Built **geospatial visualizations** using OpenStreetMap, Folium, and GeoPandas to assess route quality and geodata accuracy for an urban mobility application.

Research Intern

Apr 2024 – Aug 2024

National Institute of Informatics (NII)

Tokyo, Japan

- Constructed a **synthetic dataset** of over 150,000 renderings using Blender and Objaverse, covering 12 viewpoints and 5 HDR illumination environments per object, with paired depth and surface-normal maps, to supervise joint novel-view synthesis and relighting from a single image.
- **Fine-tuned a pretrained latent diffusion model** to jointly **synthesize novel views and relight objects from a single input image** conditioning on source illumination and viewpoint; analyzed failure modes in disentangling geometry, appearance, and illumination from a single observation.

Research Assistant

Fraunhofer IZFP, Project [SmartPigHome](#)

Aug 2023 – Mar 2024

Saarbrücken, Germany

- Supported **computer-vision experiments** for livestock monitoring, including object detection and motion-based video analysis from camera feeds.
- Developed an **audio-annotation tool** for veterinary recordings that replaced a manual paper-based workflow, reducing annotation time by up to 80%.

Research Assistant

German Research Center for Artificial Intelligence (DFKI), Project KIMonoS

Jun 2023 – Mar 2024

Saarbrücken, Germany

- Designed a **scheduling algorithm for dynamic multi-vehicle shuttle routing** and evaluated its behavior under varying passenger demand through large-scale traffic simulations using SUMO.

Computer Vision Intern

Aurify Systems Pvt. Ltd.

Jan 2022 – Apr 2022

Mumbai, India

- Developed end-to-end **machine-vision pipelines for retail and warehouse environments**, covering product detection, empty-shelf monitoring, and CCTV-based activity analysis, deployed under resource constraints using CNN-based detection pipelines.
- Built a human **pose-based activity-recognition system** using a two-stage pipeline of pose estimation and SVM classification for real-time suspicious-activity detection, achieving 91% accuracy in field evaluations.

Market Research Analyst

Finbits India

Aug 2020 – Sep 2021

Mumbai, India

- Applied **data-driven financial analysis** on company fundamentals to assess stock attractiveness, and translated quantitative insights into **written investment commentaries**.

Projects

- **Hierarchical VAE for Image Compression.** Implemented a **hierarchical extension of a VAE-based image-compression model**, improving reconstruction quality by approximately 5 dB PSNR over JPEG at comparable bitrates through architectural modifications. [[report](#)]
- **Bird Call Recognition.** Built an end-to-end **audio classification pipeline to identify bird species** from field recordings, applying MFCC feature extraction to isolate call signals from noisy environmental captures, and training a CNN classifier on the resulting spectral representations; attaining 82% test accuracy.

Technical Skills

Programming: Python, SQL, R, C++

Mathematics & Statistics: Probability Theory, Statistical Methods, Time-Series Analysis

Deep Learning: PyTorch, Transformers, Diffusion Models, VAEs, CNNs, Multi-GPU & Distributed Training

ML & Data: scikit-learn, pandas, NumPy, Hugging Face (Transformers, Diffusers), Interpretable ML, ML Efficiency (Pruning, Quantization), Deep Learning Privacy (Membership Inference, Differential Privacy)

Infrastructure & Tools: SLURM, Docker, Git, Weights & Biases, Tableau, Blender

Spoken Languages: English, Hindi, Marathi, German (A2, progressing toward B1)

Student Leadership

- **Executive Member, Students' Council**, Colloquium (Management Cell), NMIMS Aug 2018 – Jul 2019
Organized student events on management, finance, and entrepreneurship.
- **Sub-head of Photography**, Social Conclave, NMIMS Jan 2020
Led the photography team for covering a student forum on social and economic issues.

Volunteering

- **Teaching Volunteer**, The Satsang Foundation. Tutored students from the tribal community of Aarey Colony, Mumbai, in Physics, Mathematics, and English.
- **COVID-19 Relief Volunteer**, The Satsang Foundation. Supported food distribution for economically affected communities during the pandemic.